

Midterm Bonus Assignment

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Abstract

This document serves as the solution to the Midterm Bonus Assignment problems.

1 Introduction

Euler showed that the sum of the inverses of the squares of the positive integers converges to $\pi^2/6$:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The purpose of this assignment is to prove this fact using methods from single and multivariable calculus.

2 Problem 1

Prove that the double integral and the infinite series shown below are equal:

$$\int_0^1 \int_0^1 \frac{1}{1-xy} dx dy = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

For the double integral, we have:

$$\int_0^1 \int_0^1 \frac{1}{1-xy} dx dy = \int_0^1 [\ln(1-xy)]_0^1 dy = \int_0^1 \ln(1-y) dy = [-y \ln(1-y) + y]_0^1 = 1.$$

Expanding the integrand into an infinite series, we have:

$$\frac{1}{1-xy} = \sum_{n=0}^{\infty} (xy)^n = \sum_{n=0}^{\infty} x^n y^n.$$

3 Conclusion